

Formula for Efficient Unemployment: $u^* = \sqrt{uv}$

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Social planner's problem

- Allocate labor across producing / recruiting / jobposting to maximize welfare.

$$\min u + v \quad \text{subject to} \quad v = v_0 / u$$

$$\min_u u + v_0 / u$$

— convex minimization problem

— FOC is necessary & sufficient for global minimum.

$$\bullet \text{ FOC: } \frac{\partial (u + v_0/u)}{\partial u} = 0$$

$$\Leftrightarrow 1 - v_0/u^2 = 0$$

$$\Leftrightarrow u^2 = v_0$$

$$\Leftrightarrow u^* = \sqrt{v_0}$$

location of Beveridge curve:

$$u v = v_0$$

$$\Leftrightarrow u^* = \sqrt{u v}$$

Efficient unemployment rate is the geometric
average of unemployment rate & vacancy rate